Labwork 2 - Introduction to cryptography

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Write a program in Matlab, C++ or python to solve below exercises.

Exercise 1

Given two numbers: $a = 52698$, $b = 61430$, compute great common divisor of a and b.

Exercise 2

Given a set of numbers

$$A = \{290345, 218585, 143231, 164172, 155768, 423151, 239707, 153544, 287390, 480837\}$$

find all pairs of number that are relatively prime to each other.

Exercise 3

Based on the provided description of the square- and-multiply algorithm, calculate exponentiation:

$$x^e \ mod \ m$$

Compute:

$$856^{25} \ mod \ 7$$

Show the manual computation using the algorithm and compare it to returned results from your program.

Exercise 4

Find multiplicative inverse of $x = 65537$ in the modulo $N = 35256$.